

# Sinan Muhammed

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## EXPERIENCE

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- **Bud Ecosystem** New York, US · Remote  
*Staff Engineer* Sept 2025 – Present
  - Wrote K8s operator to optimize startup time (40 - 400% performance gains on production LLM workloads)
  - Implemented fault tolerant, high availability databases, storage solutions, and automated backup systems.
  - Identified and remediated vulnerabilities, Code Reviews, Ensure project follows industry best practices.
  - Integrated application and cluster telemetry/monitoring via OpenTelemetry, and Automation pipelines.
  - Led complete Infrastructure as Code (IaC) migration (Terraform, BareMetal, NixOS, K8s, VXLAN, BGP).
  - Built various internal tools to improve overall developer experience and efficiency.
- **NocoDB** San Francisco, US · Remote  
*Development and Operations Engineer* Sept 2024 – Sept 2025
  - Built a Managed Hosting Platform for NocoDB using Typescript, Nix, and AWS.
  - Led the AWS Infrastructure as Code (IaC) Transition using Terraform and Nix.
- **Google Summer of Code (GSoC)** Mountain View, CA · Remote  
*Software Development* May 2024 – Sept 2024
  - Research and Build C software for Precedence Constrained Knapsack (PCKP, scheduling optimization).
  - Profile and optimize software to meet production performance requirements (pthreads, custom hash table).
  - Write tests using Meson (Unit Tests, C) and NixOS testing framework (Integration Tests, Nix).
- **NixOS** Remote  
*Open Source Software Maintainer* Nov 2023 – Present
  - Fixed various bugs in NixOS networking stack (DNS, Interfaces).
  - Write and maintain tests, packages, and NixOS Modules.

## PROJECTS

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- **namescale** Go, Nix  
*Zeroconf Wildcard DNS for Tailscale*
  - Headscale compatibility.
- **redq** C, Go, Linux eBPF, Nix, SQL, TypeScript  
*Network observability and firewall using eBPF*
  - Bandwidth and session based usage tracking per device, activity detection.
  - DNS filtering, and MAC filtering.
- **mpvsub** Lua  
*Automatic subtitles for mpv*
  - Purpose made fuzzy matching algorithm.
  - No API keys, or accounts needed, native implementation with minimal dependencies.
- **npass** C, Nix, D-Bus, SQL  
*A passwordstore and Freedesktop.org Secret Service API implementation*
  - Backward compatibility with existing unix passwordstore databases.
- **ctfbox** LXC/Containers, Lua, SSH, Nix/NixOS  
*Declarative Linux environments for Cochin University Dhishna CTF*
  - Automated thin GNU/Linux environment provisioning with user sign up.

## CERTIFICATIONS

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- **Hurricane Electric IPv6 Certification** Feb 2024  
<https://www.he.net/>

## EDUCATION

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- **Cochin University of Science and Technology** Kochi, India  
*Bachelor's Degree in Computer Science* Sep 2021 – Jul 2025